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Inter-lead correlation analysis for automated detection of cable reversals in 12/16-lead ECG

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ABSTRACT

Background and objective: A crucial factor for proper electrocardiogram (ECG) interpretation is the correct electrode placement in standard 12-lead ECG and extended 16-lead ECG for accurate diagnosis of acute myocardial infarctions. In the context of optimal patient care, we present and evaluate a new method for automated detection of reversals in peripheral and precordial (standard, right and posterior) leads, based on simple rules with inter-lead correlation dependencies.

Methods: The algorithm for analysis of cable reversals relies on scoring of inter-lead correlations estimated over 4s snapshots with time-coherent data from multiple ECG leads. Peripheral cable reversals are detected by assessment of nine correlation coefficients, comparing V6 to limb leads: (I, II, III, –I, –II, –III, –aVR, –aVL, –aVF). Precordial lead reversals are detected by analysis of the ECG pattern cross-correlation progression within lead sets (V1–V6), (V4R, V3R, V3, V4), and (V4, V5, V6, V8, V9). Disturbed progression identifies the swapped leads.

Results: A test-set, including 2239 ECGs from three independent sources—public 12-lead (PTB, CSE) and proprietary 16-lead (Basel University Hospital) databases—is used for algorithm validation, reporting specificity (Sp) and sensitivity (Se) as true negative and true positive detection of simulated lead swaps. Reversals of limb leads are detected with Se = 95.5–96.9% and 100% when right leg is involved in the reversal. Among all 15 possible pairwise reversals in standard precordial leads, adjacent lead reversals are detected with Se = 93.8% (V5–V6), 95.6% (V2–V3), 95.9% (V3–V4), 97.1% (V1–V2), and 97.8% (V4–V5), increasing to 97.8–99.8% for reversals of anatomically more distant electrodes. The pairwise reversals in the four extra precordial leads are detected with Se = 74.7% (right-sided V4R–V3R), 91.4% (posterior V8–V9), 93.7% (V4R–V9), and 97.7% (V4R–V8, V3R–V9, V3R–V8). Higher true negative rate is achieved with Sp > 99% (standard 12-lead ECG), 81.9% (V4R–V3R), 91.4% (V8–V9), and 100% (V4R–V9, V4R–V8, V3R–V9, V3R–V8), which is reasonable considering the low prevalence of lead swaps in clinical environment.

Conclusions: Inter-lead correlation analysis is able to provide robust detection of cable reversals in standard 12-lead ECG, effectively extended to 16-lead ECG applications that have not previously been addressed.

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1. Introduction

The standard 12-lead electrocardiogram (ECG) is a routine non-invasive clinical investigation for the management of acute and chronic cardiovascular diseases [1]. It is sensitive to anterior, inferior and left-lateral projection of the cardiac electrical activity [2], as illustrated in Fig. 1. Although the redundancy of information contained in different perspectives from multiple leads is used to improve recognition of ECG abnormalities [1], there are areas of the heart that are not directly assessed by a 12-lead ECG, and thus posterior and right ventricular acute myocardial infarctions (AMI) are likely to be misdiagnosed, which might critically delay the early intervention and treatment [3]. In patients with ischemic-like chest pain, the use of additional leads might not only confirm the presence of AMI, but also provide a more accurate reflection of the true extent of myocardial damage [4,5]. Four additional right precordial leads (V3R, V4R, V5R, V6R), placed as a mirror image to the standard precordial electrode positions, are administered in the clinical setting for screening the right ventricle for diagnosis of right ventricular AMI in patients with inferior infarction [1,6,7]. Standards recommend detection of ST-segment elevation >0.5 mm in leads V3R and V4R [8], reported to be moderately sensitive and specific for right ventricular injury (V4R: sensitivity Se = 88–100%, specificity Sp = 78–82%, 83–92% diagnostic accuracy [4]). Examination of three additional posterior chest leads (V7, V8, V9) on the same horizontal plane as V6 gives a view to the posterior wall of the left ventricle and has been administered for diagnosis of posterior AMI associated with occlusion of the circumflex artery or dominant right coronary artery, not always seen in the standard 12-lead ECG [4,9–12]. ST-segment elevation >0.5 mm in posterior leads V8 and V9 is sensitive for posterior wall infarction [8] (as high as 90%, with predictive accuracy up to 93.8% [4,7]). Due to the distance of the heart (which is more anterior in the chest), the voltages recorded in the posterior leads are often small in amplitude [4,7], which requires a high-resolution ECG recording system.

An important problem in the recording of multilead ECGs is the improper placing of electrodes on the body [13], considering relatively frequent human mistake rates of: 36% for precordial electrode misplacement with more than 3.2 cm from the proper anatomical landmarks [14]; 0.8% and 7.5% for limb lead interchanges in 12-lead ECG and Holter devices, respectively [15]; and 0.4% and 4% for 12-lead ECG reversals in clinical and intensive care settings, respectively [16]. The proportion of inadvertent electrode reversals is suggested to be higher for the non-routine 16-lead ECG in an emergency setting: 5%, 8%, 8% and 19% for limb leads, standard chest leads, posterior V8–V9 leads and right precordial V3R–V4R leads, respectively [17]. Electrode placement errors can produce prominent changes in the ECG waveform morphology [18–20] and pathological patterns mimicking ectopic atrial rhythm, bundle branch blocks, ventricular hypertrophy, ventricular pre-excitation, dextrocardia, simulation or concealing of myocardial ischemia or infarction [21–27] that might generally result in incorrect data interpretation, erroneous diagnosis and lack of proper therapy if not opportunely corrected [1,8]. The REVERSE mnemonic has been suggested as a useful tool that could be easily followed by the medical staff to identify common electrode placement errors produced by typical abnormal ECG patterns [27], and ECG interpretation in the context of the patient's history is advised [22]. Confirmation with a second ECG is usually required, although less common lead placement errors are often difficult to be detected and might remain concealed. In the context of the optimal patient care, saving of money, time and valuable data, the *recent recommendations on ECG standardization and interpretation advise* training of the medical staff on the avoidance of lead switches and guidelines for their recognition, as well as incorporation of automated lead-switch detection algorithms into digital ECG devices [1].

Extended reviews of the most common cases for 12-lead ECG reversals reveal their typical effect on the P-QRS-T morphology alteration together with some basic principles for recognition of different lead reversals [21,22,24,25,27]. The implementation of these principles into automated lead-switch

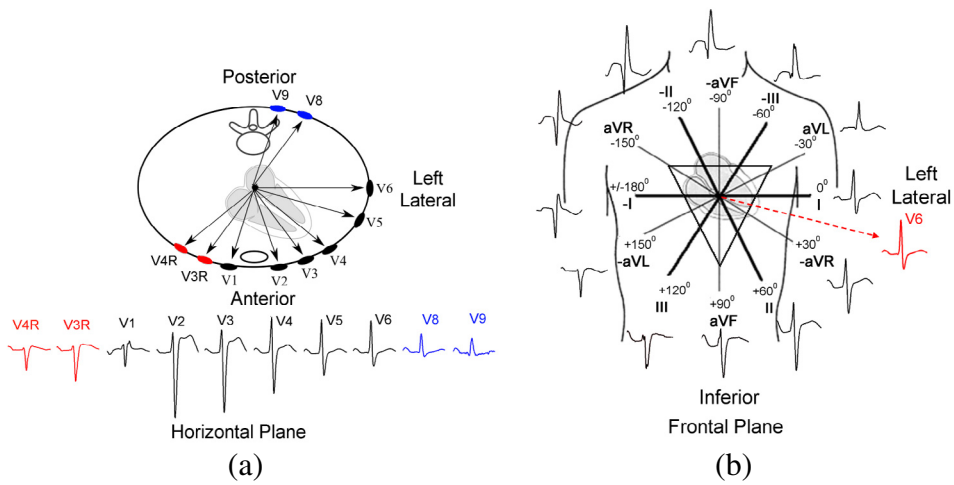

Fig. 1 – Acquisition of cardiac electrical activity via 16-lead ECG. The inter-lead normal PQRST progression is shown in (a) 10 precordial leads on the horizontal plane and (b) 6 standard limb leads (positive and inverted projection) on the frontal plane, compared to V6.

detection algorithms is a challenging task that has been extensively studied for limb leads. The published algorithms detect left arm (LA), left leg (LL) and right arm (RA) reversals by morphological analysis of the P wave (P wave amplitude and polarity [28,29], frontal P axis [29–33], direction of inscription of the P loop [29,32]), the QRS complex (QRS amplitude and polarity [29,34], frontal QRS axis [30,31,33,35], QRS loop rotation [29]), as well as using the redundancy of information in the 12-lead ECG by reconstruction and correlation techniques [33,36], redundancy features [29] and comparison between a composite lead aVF/I and V6 [34,37]. The swaps, involving the neutral electrode on the right leg (RL), are easily captured by a flat line in any of the leads I, II or III [34,38], except the not detectable RL–LL swap for which the ECG remains unchanged [39]. The overall detection accuracy for limb lead reversals is relatively high—published automated algorithms provide Sp > 99% and Se > 87%. An exception is the LA–LL swap, which is obviously the most difficult to detect (18% [36], 58% [30], up to 90% [28]), although it has been reported to be the most frequently seen reversal in 12-lead ECG (58% of all lead errors) [15].

The automated detection of precordial lead reversals is elaborated in fewer studies, using only ECG morphology [30], ECG morphology and redundancy information [29], correlation between any lead and its reconstruction model [36], PQRST pattern correlations and time-amplitude alignments of S, R peaks from V1 to V6 [34,40]. Compared to limb leads, the identification of precordial lead reversals keeps also Sp > 99%; however, the pairwise lead reversals are less accurately detected with Se range of 78%–100% [34,36,40], and even lower Se = 45%–83% [30] and Se = 17%–86% [29]. These results keep open the room for improvement of computer-based algorithms for detection of all kinds of lead reversals that would benefit the quality of health care.

The use of 16-lead ECG for accurate AMI diagnosis in patients with ischemic-like chest pain, taking additional views of the right ventricle from right precordial leads (VR3, VR4) and the posterior left ventricular wall from posterior leads (V8, V9), requires additional education in placement of the extra chest electrodes, ECG reading and interpretation by the human expert, as well as by computer-based ECG analysis. We aim to present a clinically relevant method for automated detection of 16-lead ECG based on inter-lead correlations and to evaluate its unbiased performance on independent proprietary and public ECG databases. The possible variations in ECG electrode placement are enormous [21]; therefore, this article will address reversals within the following cable sets: four limb electrodes (RA, LA, RL, LL), six standard chest leads (V1–V6) and four extra chest leads (V4R, V3R, V8, V9), the latter being not routinely used, thus presenting a challenge to the medical staff for correct placement of electrodes and visual identification of reversals, which to our knowledge have not still been evaluated by any automated algorithm.

2. ECG databases

This study is performed with resting ECG recordings from three independent sources—a clinical proprietary 16-lead ECG database from the Basel University Hospital (BUH), and two publicly available 12-lead ECG databases, including the Physionet Physikalisch-Technische Bundesanstalt (PTB) diagnostic ECG database [41,42] and the Common Standards for Electrocardiography (CSE) diagnostic database [43]. We consider for further analysis only one 4s segment per recording (fixed for all recordings from the 4th to 8th second to prevent from possible pre-processing border effects), nevertheless the different duration of the raw ECG data within databases. All recordings are visually inspected for RA–LA interchange for rejection of cases affected by this frequent type of cable reversal (easily recognizable during sinus rhythm by the presence of negative P and QRS waves in lead I, never observed in healthy hearts and very unusual in the presence of cardiac disease [21]). Since other reversals could not be identified with such confidence, the remaining ECG signals are assumed to be correctly acquired in the clinical environment.

General characteristics of the used databases are summarized in Table 1. Specific details are provided below:

The BUH database contains resting 16-lead ECG recordings, collected from 1315 chest pain patients at the Emergency Department of the Basel University Hospital with unavailable diagnosis. The signals are acquired via a CS-200 Excellence device (Schiller AG, Switzerland) in the period January 2012–December 2014. The device does not give any feedback to the user with respect to any lead reversal. The BUH database has been divided in two independent subsets from non-overlapping periods—BUH (Training set) from 2012–2013 and BUH (Test set 1) from 2014. The standard 12-lead ECGs are taken from the Physionet PTB diagnostic database, originally containing 549 ECG recordings from 290 subjects (52 healthy controls, 148 patients with myocardial infarction and 90 persons suffering from other cardiac diseases). Five records have been excluded from further analysis as they contain unacceptable errors—four recordings ("s0197lre," "s0196lre," "s0236lre," "s0166_re") are indicative for RA–LA reversal, one recording ("s0544_re") has unacceptable quality with spike artifacts and no visible ECG waveform in any lead. Thus a number of 544 ECG recordings are further referred as the PTB (Test set 2). The standard 12-lead ECGs, which are taken from the CSE diagnostic database (also called CSE data set 5), include original ECGs from 382 normal rhythms, 219 ECGs from patients with hypertrophy, 547 ECGs representing myocardial infarction and 72 ECGs from persons suffering from other cardiac diseases. All 1220 recordings are further referred as the CSE (Test set 3).

3. Methods

3.1. *Conception for detection of 12/16-lead ECG reversals*

A two stage concept for detection of reversals in standard 12-lead ECG and its further expansion to 16-lead ECG is followed according to Fig. 2. The first stage checks for reversals in standard precordial leads (V1–V6), and only if specific leads could be considered with confidence they are used as a reference for a second stage verification of three independent electrode sets: limb electrodes (RA, LA, LL, RL) vs. (V6), right precordial leads (V3R, V4R) vs. (V3, V4), and posterior leads (V8, V9) vs. (V4, V5, V6). All algorithms for analysis of electrode reversals rely on scoring of inter-lead correlation coefficients (Eq. 1) estimated

Table 1 – Summary of all ECG databases, used either for algorithm development (Training set) or validation (Test sets 1, 2, 3).

Datasets	Nb analyzed recordings	Total duration	Leads	Sampling frequency	Ampl. resolution
BUH (Training set)	840	5 min	16-leads I, II, II, aVR, aVL, aVF, V1–V6, V8, V9, V3R, V4R	1 kHz	1 μV
BUH (Test set 1)	475	5 min	16-leads I, II, II, aVR, aVL, aVF, V1–V6, V8, V9, V3R, V4R	1 kHz	1 μV
PTB (Test set 2)	544	32–120 s	12-leads I, II, II, aVR, aVL, aVF, V1–V6	1 kHz	0.5 μV
CSE (Test set 3)	1220	10 s	12-leads I, II, II, aVR, aVL, aVF, V1–V6	500 Hz	<5 μV

over several seconds of time-coherent data from multiple leads, pre-filtered in a bandwidth (1–30) Hz. No special requirements for P-QRS-T patterns delineation are applied, considering that the high-amplitude components of the ECG signal (i.e. QRS complexes, T-waves) would have the greatest impact on the estimated cross-correlation coefficients between leads:

$$r(L1, L2) = \frac{\sum_t L1_t L2_t}{\sqrt{\sum_t L1_t^2 \sum_t L2_t^2}} \quad (1)$$

where t represents the time series of leads L1 and L2.

The analytical approaches defined in further subsections are based on observations over the 16-lead BUH database (Training set).

3.2. *Detection of precordial lead reversals*

The method for detection of precordial lead reversals estimates the inter-lead ECG waveform similarities, following the normal progression of the cardiac vector projection on the horizontal plane, starting from the right chest leads (V4R, V3R), through the anterior leads (V1, V2), the septal leads (V3, V4), the left lateral leads (V5, V6), to the posterior leads (V8, V9) – see Fig. 1a. Specific lead reversals are detected by correlation analysis within three precordial lead sets:

- Set for standard precordial leads: SPL = [V1, V2, V3, V4, V5, V6]. A square matrix rSPL(6 × 6) is defined to contain the cross-correlation coefficients according to Eq. 1, by substituting L1 and L2 with all pairwise combinations of leads in SPL. An illustration is provided in Fig. 3.
- Set for right precordial leads: RPL = [V4R, V3R, V3, V4]. Although leads V1 and V2 are anatomically close to leads V4R and V3R, they are not used in the RPL set because we found that their correlations with V4R and V3R might show a disturbed progression of the ECG waveform for the correct electrode order. This finding is a consequence from the reported great probability that V1 and V2 are incorrectly positioned in the second or third intercostal space in

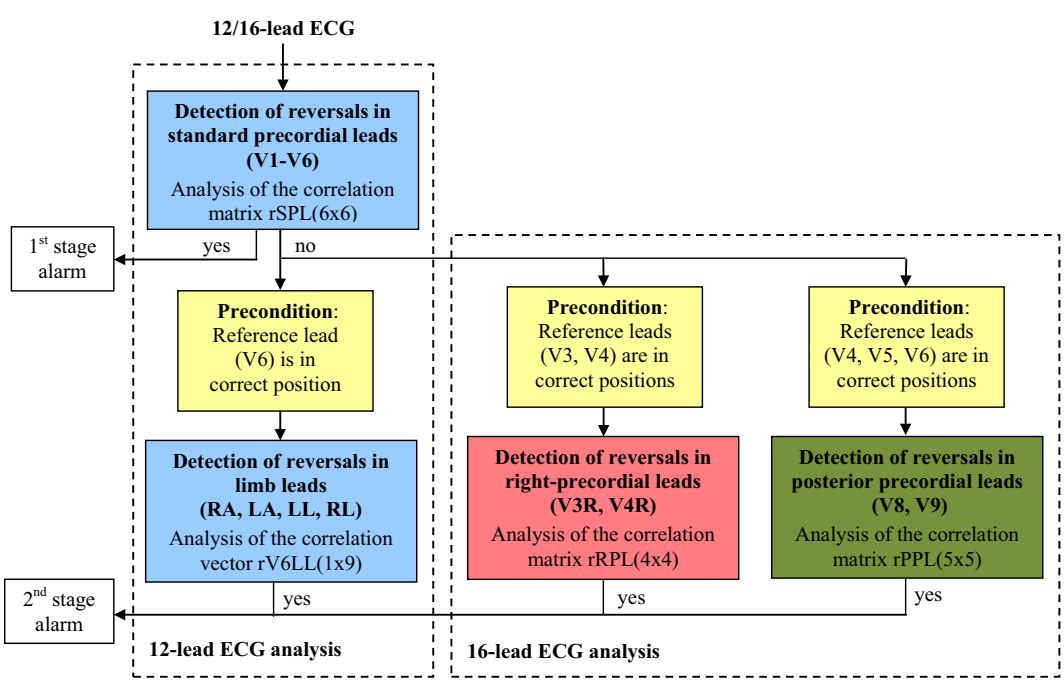

Fig. 2 – Two-stage concept diagram for detection of reversals in 12-lead or 16-lead ECG. If the first stage detects a reversal in any of the standard precordial leads (V1–V6), then an alarm for lead reversal is activated and the second-stage processing is disabled.

	V4R	V3R	V1	V2	V3	V4	V5	V6	V8	V9
V4R	1	>			>	>				
V3R	<	1			>	>				
V1			1	>	>	>	>	>		
V2			<	1	>	>	>	>		
V3	<	<	<	<	1	>	>	>		
V4	<	<	<	<	<	1	>	>	>	>
V5			<	<	<	<	1	>	>	>
V6			<	<	<	<	<	1	>	>
V8						<	<	<	1	>
V9						<	<	<	<	1

Fig. 3 – Illustration of the square correlation matrices defined for the three lead sets of 10 precordial leads: rSPL(6 × 6) – blue rectangle enclosing V1 to V6; rRPL(4 × 4) – red cells including V4R, V3R, V3, V4; rPPL(5 × 5) – green cells including V4 to V9. The signs should be interpreted in the following context for the correct lead placement: "1" – in the diagonal cells, showing the maximal correlation when a lead is compared to itself; ">" – in the cells above the diagonal, corresponding to the gradual decrease of the correlation coefficients rightwards from the diagonal when inter-lead distance increases; "<" – in the cells below the diagonal, corresponding to the gradual decrease of the correlation coefficients leftwards from the diagonal when inter-lead distance increases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

emergency care [13], with superior quadrant displacement of more than 1.6 cm in more than 50% of routine applications [14]. Besides, studies of body surface maps show that lead V2 is the most sensitive to displacement errors that inherently arise from the physical proximity to the signal source [18,19]. Analogically to rSPL, a square matrix rRPL(4 × 4) is defined to contain the cross-correlation coefficients between all pairwise combinations of leads in RPL (Fig. 3).

- Set for posterior precordial leads: PPL = [V4, V5, V6, V8, V9]. A relevant square matrix rPPL(5 × 5) is defined to contain the cross-correlation coefficients between all pairwise combinations of leads in PPL (Fig. 3).

The matrices rSPL(6 × 6), rRPL(4 × 4) and rPPL(5 × 5) are further analyzed by comparison of the correlation coefficients over the rows within each matrix. Any deviation from the trend for gradual increase toward the diagonal entries, where a lead is compared to itself, is considered to be due to a lead reversal as announced in our preliminary studies for rSPL(6 × 6) [40], rRPL(4 × 4), and rPPL(5 × 5) [44]. This assumption is illustrated in Fig. 1a, showing the typical ECG pattern progression along the precordial leads, where (i) a gradual increase of the amplitude ratio R-wave/S-wave is observed from V4R to V6 that influences the cross-correlations in rSPL(6 × 6) and rRPL(4 × 4); and (ii) a gradual decrease of the S-wave amplitude from V4 to V9 and appearance of Q-wave in V8, increasing in V9, determine the correlation trends in rPPL(5 × 5). The method is sensitive to the type of the lead reversal, i.e. the leads involved in the distorted cross-correlation progression are suggested to be not in their correct positions.

Following the two-stage concept in Fig. 2, the analysis of the matrix rSPL(6 × 6) is only required for detection of precordial lead reversals (V1–V6) in 12-lead ECG, while analysis of matrices rRPL(4 × 4) and rPPL(5 × 5) is supplementary for identification of reversals in extra leads V4R, V3R, V8, V9 in 16-lead ECG. The detection of electrode reversals in the precordial leads is not depending on the position of the peripheral leads, although the unipolar precordial leads are acquired via the Wilson Central Terminal (WCT). When the RL neutral electrode is swapped with RA or LA, the WCT is shifted without visible distortion of the chest leads in proximity to the heart [45]. Moreover, the gradual progression of the precordial inter-lead correlations is preserved even with wrong WCT.

3.3. *Detection of peripheral lead reversals*

The detection of peripheral lead reversals, applicable in 12/16-lead ECG acquisition, has been implemented as a second stage after the standard precordial lead analysis, according to Fig. 2. The detection of RA, LA, and LL electrode swaps is based on assessment of the similarity between the limb leads and V6 – the only precordial lead that could be considered to lie in the frontal plane. Lead I is generally similar to V6 with respect to the morphology of the P wave and QRS direction and lead misplacement results in an important discordance between lead I and V6 [1].

The method for detection of peripheral electrode reversals is based on assessment of a vector rV6LL(1 × 9) with nine correlation coefficients, involving lead V6 as L1 and each of the following limb leads [I, II, III, –I, –II, –III, –aVR, –aVL, –aVF] as L2 in Eq. 1. The maximal correlation coefficient indicates the lead, which is the most similar to V6, expected to lie in the left inferior quadrant (0° to 60°) of the frontal plane, as shown in Fig. 1b. Fig. 4 summarizes the set of limb leads with expected maximal cross-correlation in rV6LL(1 × 9), considering the projection of the leads in both correct and reversed RA, LA, and LL positions [39]:

- RA, LA, and LL correct placement: [I, II, –aVR] are normally within 0° to 60° in the left inferior quadrant.
- RA–LA reversal: Results in flipped Einthoven's triangle at 180° horizontally around lead aVF, seen as inverted lead I, swapped II and III, swapped aVR and aVL. Thus [–I, III, –aVL] are considered within 0° to 60° in the left inferior quadrant.
- RA–LL reversal: Results in flipped Einthoven's triangle at 180° vertically around aVL, seen as replacement of lead I with inverted lead III, inverted lead II, swapped aVR and aVF. Thus [–III, –II, –aVF] are considered within 0° to 60° in the left inferior quadrant.
- LA–LL reversal: Results in flipped Einthoven's triangle at 180° vertically around aVR, seen as swapped leads I and II, unchanged aVR. Thus the lead set [II, I, –aVR] is the same as in correct RA, LA, and LL placement and principally the concept with maximal correlation to V6 is not applicable.

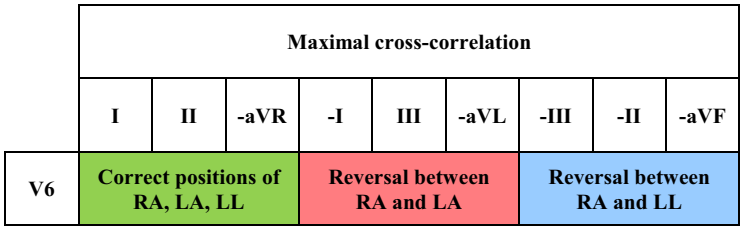

Fig. 4 – Illustration of the correlation vector rV6LL(1 × 9) showing the limb leads among which the maximal cross-correlation with lead V6 is expected for specific RA, LA, and LL positions.

Moreover, the main effects of LA–LL reversal, associated with an often subtle shift in axis and inversion of lead III, when ECG looks normal, could not reliably be detected even by a human expert without inspection of a reference ECG recording [1] or the P-wave morphology [28].

The rotations of the neutral RL electrode with RA, LA, and LL are managed as follows:

- RL–RA reversal: Results in transformation of the Einthoven's triangle to a very thin "slice" with the LA electrode at its apex, so that the difference between LL and RA potentials is small, detected as short-term voltage difference below 100 μV in lead II, as well as identical aVR and aVF.
- RL–LA reversal: Results in transformation of the Einthoven's triangle to a very thin "slice" with the RA electrode at its apex, so that the difference between LL and LA potentials is small, detected as short-term voltage difference below 100 μV in lead III, as well as identical aVL and aVF.
- RL–LL reversal: Preserves the Einthoven's triangle as the electrical signals from each leg are virtually identical. Since the ECG interpretation is kept unchanged, this reversal is not worth to be managed.

3.4. *Simulation of cable reversals*

Given that the number of all possible reversals within the full set of 12/16-lead ECG is immense, we simulate the following reduced number of the most common swaps within different cable sets:

- 10 reversals for the limb leads by transforming the limb leads vector (I, II, III, aVR, aVL, aVF) into specific configurations, as shown in the 2nd column of Table 2.
- All 15 possible pairwise reversals for the six standard chest leads (V1–V6) as listed in the 1st column of Table 3. These reversals are further referred as "one-hop" if adjacent leads are swapped (e.g. V1–V2); "two-hop" if there is one correct lead between the two swapped (e.g. V1–V3); "three-hop" if there are two correct leads between the two swapped (e.g. V1–V4); in analogy "four-hop" (V1–V5, V2–V6) and "five-hop" (V1–V6).

Table 2 – Performance for detection of limb electrodes reversals simulated for all recordings within 12/16-lead datasets.

Reversed electrodes	Simulated transforms in the standard limb leads: (I, II, III, aVR, aVL, aVF)	Train set BUH db	Test set 1 BUH db	Test set 2 PTB db	Test set 3 CSE db
None	(I, II, III, aVR, aVL, aVF)	98.7	99.2	99.3	99.7
LA–RA	(–I, III, II, aVL, aVR, aVF)	96.4	96.6	95.4	97.5
LL–RA	(–III, -II, -I, aVF, aVL, aVR)	97.1	97.5	91.7[a]	97.6
CCW rotation (RA→LL→LA→RA)	(III, –I, –II, aVL, aVF, aVR)	97.0	96.8	96.0	98.0
CW rotation (RA→LA→LL→RA)	(–II, –III, I, aVF, aVR, aVL)	97.0	97.3	91.7[a]	97.5
RL–RA	(–III, "0", III, III/2, –III, III/2)	100	100	100	100
RL–LA	(II, II, "0", –II, II/2, II/2)	100	100	100	100
CW rotation with RL (RL→RA→LA→RL)	(–II, "0", II, II/2, –II, II/2)	100	100	100	100
CCW rotation with RL (RL→LA→RA→RL)	(III, III, "0", –III, III/2, III/2)	100	100	100	100
Bilateral arm–leg rotation (RL→RA, LL→LA)	("0", –III, –III, III/2, III/2, –III)	100	100	100	100
Cross rotation (RL→RA→LL→LA→RL)	("0", –II, –II, II/2, II/2, –II)	100	100	100	100

[a] indicates significant performance differences in test vs. training set ($p < 0.05$).
Note: Sp (%) should be considered for the case when "none" electrodes are reversed (first row), while Se (%) is reported for all other reversal configurations. Each electrode reversal is simulated with specific change of the standard limb leads, shown in the order (I, II, III, aVR, aVL, aVF), where "0" means "near zero" signal, simulated as random noise below 100 μV.

Table 3 – Performance for detection of all pairwise reversals within the standard precordial lead set (V1–V6) simulated for all recordings within 12/16-lead datasets.

Pairwise reversed	Neighborhood	Training set BUH db		Test set 1 BUH db		Test set 2 PTB db		Test set 3 CSE db	
Leads		Se (%)	Sp (%)	Se (%)	Sp (%)	Se (%)	Sp (%)	Se (%)	Sp (%)
V1–V2	One-hop	97.4	98.7	97.9	99.2	96.0	98.5	97.5	98.9
V2–V3	One-hop	97.4	97.9	96.2	96.8	94.9[a]	98.2	95.8[a]	98.4
V3–V4	One-hop	97.4	99.2	96.4	98.5	96.0	99.1	95.4[a]	99.7
V4–V5	One-hop	97.3	98.8	98.1	99.2	98.7	99.3	96.7	99.1
V5–V6	One-hop	95.5	97.9	93.9	96.8	95.2	99.6[a]	92.2[a]	98.2
V1–V3	Two-hop	99.8	99.9	99.6	99.4	99.3	99.3	99.2	99.0
V2–V4	Two-hop	99.9	99.5	99.8	99.6	98.5	98.9	99.5	99.8
V3–V5	Two-hop	98.8	98.8	98.1	98.9	99.3	99.8	98.8	99.3
V4–V6	Two-hop	98.7	98.7	99.4	99.2	98.3	99.1	98.3	99.3
V1–V4	Three-hop	100	99.9	100	99.6	99.4	99.6	99.9	100
V2–V5	Three-hop	98.9	99.2	98.1	99.2	97.8	99.8	97.4[a]	99.4
V3–V6	Three-hop	99.4	98.9	98.9	98.9	99.8	99.8	100	99.3
V1–V5	Four-hop	99.9	100	99.8	99.6	99.3	100	99.8	99.9
V2–V6	Four-hop	99.3	99.3	98.5	99.2	99.3	99.8	99.3	99.4
V1–V6	Five-hop	97.0	100	94.7[a]	99.6	94.3[a]	99.8	94.3[a]	100
Mean ± std		98.4 ± 1.4	99.1 ± 0.7	98.0 ± 1.9	98.9 ± 0.9	97.7 ± 1.9	99.4 ± 0.5	97.6 ± 2.3	99.3 ± 0.5

[a] indicates significant performance differences in test vs. training set ($p < 0.05$).
Note: Sp (%) is the rate of true negative alarms indicated for the particular lead pair in the correct lead placement; Se (%) is the rate of true positive alarms indicated within the particular lead pair when that reversal is simulated.

- All 6 possible pairwise reversals for the four extra chest leads (V4R, V3R, V8, V9) as listed in the 1st column of Table 4.

Simulation of reversals in limb leads and standard chest leads has been done for all recordings in BUH (Training set), BUH (Test set 1), PTB (Test set 2) and CSE (Test set 3), while simulation of extra chest leads reversals has been done for all recordings in 16-lead BUH (Training set) and BUH (Test set 1). For each dataset, the performance is reported as sensitivity for the simulated lead reversal and specificity for the correct lead placement.

Fisher's exact test is used to study statistically significant percent point difference (PPD) of Sp and Se values for the training vs. test datasets, considering significant $p < 0.05$. Statistically insignificant PPD indicates high reproducibility of performance among databases.

4. Results

The method for detection of lead reversals is evaluated over 4s snapshots with time-coherent leads in each recording. The following is the analysis of the results concerning reversals within different cable sets:

- Four limb electrodes (RA, LA, RL, LL): The performance for detection of the correct placement and the simulated 10 electrode reversals is reported in Table 2. Sp between 98.7% (training BUH) and 99.2%, 99.3%, and 99.7% (test BUH, PTB, and CSE) indicates high Sp reproducibility with insignificant PPD ≤ 1% ($p > 0.05$). The four considered LA, RA, and LL reversals (RA–LA, LL–RA, clockwise (CW) and counterclockwise (CCW) rotations) are detected with Se from 96.4%

Table 4 – Performance for detection of all pairwise reversals within the set of four extra chest leads (V4R, V3R, V8, V9), relevant only for 16-lead ECG datasets.

Pairwise reversed	Neighborhood	Training set BUH db		Test set 1 BUH db	
Leads		Se (%)	Sp (%)	Se (%)	Sp (%)
V4R–V3R	One-hop	80.5	85.5	74.7[a]	81.9[a]
V8–V9	One-hop	91.9	93.9	91.4	91.4[a]
V4R–V9	One-hop	93.6	100	93.7	100
V4R–V8	Two-hop	97.7	100	97.7	100
V3R–V9	Two-hop	98.1	100	97.7	100
V3R–V8	Three-hop	98.5	100	97.7	100
Mean ± std		93.4 ± 6.9	96.5 ± 6.0	92.2 ± 8.9	95.5 ± 7.5

[a] indicates significant performance differences in test vs. training set ($p < 0.05$).
Note: Sp (%) is the rate of true negative alarms indicated for the particular lead pair in the correct lead placement; Se (%) is the rate of true positive alarms indicated within the particular lead pair when that reversal is simulated.

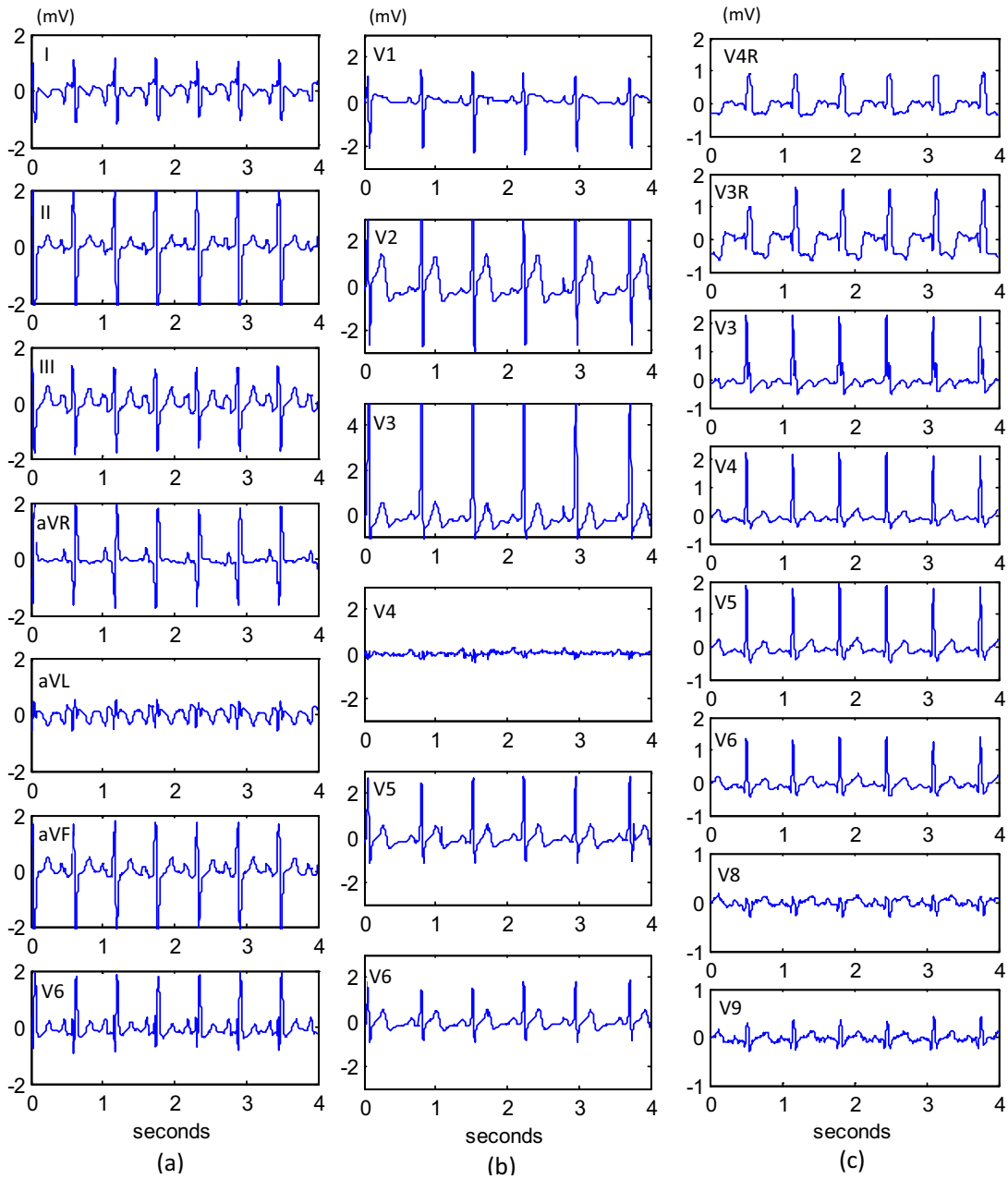

Fig. 5 – Examples of erroneously detected lead reversals that lower the specificity: (a) record "s0433_re" from PTB database with diagnosis for anterior MI: LA–RA reversal is alarmed as lead V6 is best correlated with –I (the negative P, T-waves in I and aVL are suspicious for real electrode reversal); (b) record "s0545_re" from PTB database with status "healthy control": the low amplitude in V4 triggers an alarm for lead reversal; (c) a record from BUH database: two reversals of V4R–V3R and V8–V9 are erroneously detected. We suggest that V3R and V4R are correctly placed but the identified right bundle branch block is disturbing the ECG pattern progression in these leads. We are suspicious for real reversal of V8–V9, and therefore an alarm for checking these leads is worth.

to 97.1% (training BUH), 96.6% to 98.0% (test BUH, CSE) and 91.7% to 96.0% (test PTB), denoting a high Se reproducibility with insignificant PPD ≤ 1% ($p > 0.05$), except PPD = 5.4% ($p < 0.001$) for RA–LL and CW reversals in PTB database. The latter might be due to disturbed correlation V6 vs. [–III, –II, –aVF] for some recordings in PTB database that has a prevalence of myocardial infarctions. When RL is involved in the reversals of limb electrodes, Se = 100% for all databases. An example of erroneously detected reversals that lower Sp is shown in Fig. 5a.

- Six standard chest leads (V1–V6): The performances for detection of the simulated 15 pairwise lead reversals are reported in Table 3. As soon as the method indicates the swapped leads (with distorted cross-correlation progression) we are able to calculate Se and Sp for each particular reversal. For all reversals, Sp > 97.9% (training BUH) with high

Sp reproducibility among test databases, showing statistically insignificant PPD ≤ 1.1% ($p > 0.05$), except PPD = 1.7% ($p = 0.027$) for V5–V6 reversal in PTB database. All reversals are detected with Se > 95.5% (training BUH), reproduced in the test databases with statistically insignificant PPD < 1.5%, ($p > 0.05$), except for the reversals V1–V6 (PPD < 2.7%), V2–V3 (PPD < 2.5%), V3–V4 (PPD = 2%), V5–V6 (PPD = 3.3%), and V2–V5 (PPD = 1.5%) in specific test databases, highlighted in Table 3. An example of erroneously detected reversals that lower Sp is shown in Fig. 5b.

- Four extra chest leads (V4R, V3R, V8, V9): The performances for detection of the simulated 6 pairwise lead reversals are reported in Table 4. The lowest Sp = 85.5% and Se = 80.5% (train BUH) are reported for V4R–V3R reversal with significant PPD of 3.6% and 5.8% ($p < 0.05$) in test BUH. All other pairwise reversals have Sp = 100% and Se > 91.9% (train BUH) with insignificant PPD < 0.8% ($p > 0.05$) in test BUH, except for the reversal V8–V9 with Sp = 93.3% (train BUH) and PPD = 2.5% ($p < 0.05$). An example of erroneously detected reversals that lower Sp is shown in Fig. 5c.

5. Discussion

This study presents a method for automated detection of cable reversals based on inter-lead correlations in the most in use standard 12-lead ECG, effectively extended to 16-lead ECG applications that have not previously been addressed by other authors. This paper is clinically relevant compared to our previous technical reports, which are basically focused on solving specific subtasks and limited in respect of application for incomplete set of chest leads [40,44] and validation on restricted population [34,40]. The definition of the cross-correlations within rV6LL(1 × 9) for limb leads, rSPL(6 × 6) for standard precordial leads, and [rRPL(5 × 5), rPPL(4 × 4)] for the extra precordial leads has been set on clinically relevant observations over a large training 16-lead BUH database from an emergency department. We provide performance validation on three independent test ECG databases, two of them (PTB and CSE) being common standards for inter- and intra-study comparisons. We deal with the full set of recordings in the databases, without selection or exclusion criteria based on particular disease types. The observed high reproducibility of the performance among databases (insignificant PPD < 1.5% in prevalent number of tests) proves that the proposed simple model is sufficient to account for variances in the data and is not overfitted to the training data.

In clinical practice, the requirements toward automated lead-switch detection algorithms give a preference to low false positive alarms that would not disturb and provoke the medical staff to frequently discard or even deactivate the automated decision. The presented algorithm fulfills these requirements with a mean Sp > 99% for detection of reversals in standard 12-lead ECG (Tables 2 and 3), considering the low prevalence of such interchanges when medical personnel is routinely trained to recognize them. The lower specificity is, however, not critical to alarm reversals in extra precordial leads (Table 4): right-sided swaps V4R–V3R (Sp > 81.9%), posterior swaps V8–V9 (Sp > 91.4%) and none false positive swaps between right and posterior leads (Sp = 100%), considering that clinicians are not yet routinely aware of electrode positioning, reading and interpreting the additional views from 16-lead ECG, especially for AMI diagnosis in emergency care. We show examples to illustrate some typical reasons for false positive alarms, such as PQRST pattern changes that mimic real electrode reversal (Fig. 5a), low quality leads (Fig. 5b), and influence of cardiac disease (Fig. 5c).

The sensitivity of the algorithm to detect 12-lead ECG reversals could be highly rated among the published studies. Considering the mean Se over all test datasets, the common reversals of the three limb electrodes LA, RA, and LL are detected with a rate between 95.5% and 96.9%, going up to 100% when RL is involved in the reversal (Table 2). In comparison, the publicly reported Se ranges from 87% to 99.1% [29,31,33,36]. The detection of standard precordial lead reversals is most difficult to be managed for one-hop reversals of anatomically closed leads (Table 3). The limited mean test Se of 93.8% (V5–V6), 95.6% (V2–V3), 95.9% (V3–V4), 97.1% (V1–V2), and 97.8% (V4–V5) is due to the undetected interruption of the ECG pattern progression in adjacent leads, expected from human error electrode misplacements from the proper anatomical landmarks, as well as from the minor difference between waveforms in anatomically close electrode positions. The latter might not be clinically important as soon as the final diagnosis would not change. The mean Se goes up in the range 97.8–99.8% when two- to four-hop pairwise lead reversals are tested, which confirms that the inter-lead correlations in precordial leads are robust to different cardiac disease. The Se of this relatively simple approach is comparable and even outperforming other published automated methods, based on similar or more complex procedures. The last includes fiducial points detection, P-QRS-T morphology features measurements, reconstructed leads, fed to complex classifiers (support vector machines [29], artificial neural networks [30], decision tree [36]), and report Se in the range: 25.1–94.0% (V5–V6), 35.5–87.4% (V2–V3), 77.5–97.6% (V3–V4), 80.6–94.3% (V1–V2), and 16.9–99.1% (V4–V5) for the one-hop reversals, and 76.6–93.0% for the two-hop reversals. We could spot the most challenging detection of pairwise reversals in one-hop V5–V6 and V2–V3 and the five-hop V1–V6 (Se = 94.5%). The latter is due to the specific rSPL(6 × 6) analysis (Fig. 3), where the ones in the border diagonal entries r(V1, V1) and r(V6, V6) could not be compared to other inter-lead correlations leftwards and rightwards, when 12-lead ECG is only available. The simulated 15 possible pairwise reversals disturb the correlation progression in only 2 elements of the rSPL(6 × 6) matrix, and therefore we expect better sensitivities when more than two leads are interchanged.

Compared to the standard precordial leads, the sensitivity of the algorithm to detect pairwise reversals in extra precordial leads (Table 4) is lower for the one-hop lead couples V4R–V3R (74.7%), V8–V9 (91.4%), and V4R–V9 (93.7%), primarily due to their waveform similarity. The first two are the most problematic, as soon as the physical distance to the signal source could give similar views of the right ventricle from VR3 and VR4 and of the posterior left ventricular wall from V8 and V9. All other pairwise extra-chest leads reversals are reliably detected with Se = 97.7%.

The assumption for correct ECG acquisition in clinical environment is a pre-condition to reduce both Se and Sp for detection of cable reversals in all lead sets. Considering the high Sp (about 99%) reported for the standard 12-leads (Tables 2 and

3), we could presume that the possible influence of real reversals is negligible for this routinely applied cable configuration. However, the lower specificity for reversals of adjacent leads V4R–V3R (Sp = 81.9%) and V8–V9 (Sp = 91.4%) suggests about higher probability of human errors due to lack of experience in managing this uncommon cable configuration with extra precordial leads. The retrospective visual inspection of the waveform progression in the chest leads (standard and extended set), however, could not be used for confident verification of electrode interchange.

The present study validates the application of the interlead correlation analysis algorithm, part from the lead quality monitoring library (Schiller AG, Switzerland), for robust detection of 12/16-lead ECG reversals during the first seconds of signal acquisition. Thus a feedback could warn in time the medical staff to correct the problem, preventing both ECG readers and conventional ECG interpretation programs from misdiagnosis and improper treatment. Due to the great variety of cardiac diseases represented in the three large test ECG databases (BUH from chest pain patients, diagnostic PTB, diagnostic CSE), the reported performance can be considered very similar to the one expected in clinical environment.

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